

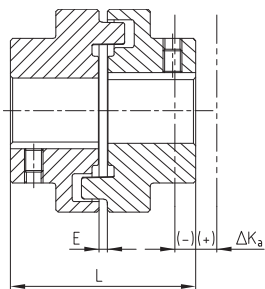
Technical data

POLY-NORM® Technical data													
Size	Torque [Nm]			Max. speed [rpm] with v = 35 m/s	Torsion angle with		Torsion spring stiffness C dyn. [Nm/rad]				Max. perm. displacement [mm] ¹⁾		
	Rated torque T _{KN}	Maximum torque T _K max	Vibratory torque T _{KW}		T _{KN}	T _K max	1.0 T _{KN}	0.75 T _{KN}	0.5 T _{KN}	0.25 T _{KN}	Axial ΔK _a	Radial ΔK _r	Angular ΔK _w
28	40	80	16	9650	4.5	6.0	0.52x10 ⁴	0.332x10 ⁴	0.187x10 ⁴	0.09x10 ⁴	± 1.0	0.20	1.2
32	60	120	24	8550			0.782x10 ⁴	0.499x10 ⁴	0.282x10 ⁴	0.135x10 ⁴	± 1.0	0.25	1.4
38	90	180	36	7650			1.35x10 ⁴	0.864x10 ⁴	0.489x10 ⁴	0.234x10 ⁴	± 1.0	0.25	1.5
42	150	300	60	6950			2.63x10 ⁴	1.68x10 ⁴	0.947x10 ⁴	0.453x10 ⁴	± 1.0	0.25	1.7
48	220	440	88	6300	4.0	5.5	2.99x10 ⁴	1.91x10 ⁴	1.08x10 ⁴	0.516x10 ⁴	± 1.5	0.30	1.8
55	300	600	120	5650			3.85x10 ⁴	2.46x10 ⁴	1.39x10 ⁴	0.664x10 ⁴	± 1.5	0.30	2.0
60	410	820	164	5150			6.76x10 ⁴	4.31x10 ⁴	2.32x10 ⁴	1.17x10 ⁴	± 1.5	0.30	2.2
65	550	1100	220	4750			8.18x10 ⁴	5.22x10 ⁴	2.7x10 ⁴	1.41x10 ⁴	± 1.5	0.35	2.4
75	850	1700	340	4200	2.5	3.5	12.29x10 ⁴	7.84x10 ⁴	4.06x10 ⁴	2.12x10 ⁴	± 1.5	0.40	2.7
85	1350	2700	540	3650			24.31x10 ⁴	15.51x10 ⁴	7.49x10 ⁴	4.19x10 ⁴	± 1.5	0.40	3.0
90	2000	4000	800	3300			36.16x10 ⁴	23.07x10 ⁴	11.14x10 ⁴	6.24x10 ⁴	± 1.5	0.45	3.4
100	2900	5800	1160	2950			54.82x10 ⁴	34.98x10 ⁴	16.89x10 ⁴	9.46x10 ⁴	± 3.0	0.50	3.9
110	3900	7800	1560	2650	1.5	2.1	79.23x10 ⁴	50.55x10 ⁴	24.4x10 ⁴	13.67x10 ⁴	± 3.0	0.60	4.3
125	5500	11000	2200	2350			102.3x10 ⁴	65.28x10 ⁴	31.52x10 ⁴	17.65x10 ⁴	± 3.0	0.60	4.8
140	7200	14400	2880	2100			164x10 ⁴	104.7x10 ⁴	50.85x10 ⁴	28.3x10 ⁴	± 3.0	0.60	5.5
160	10000	20000	4000	1900			209.1x10 ⁴	133.4x10 ⁴	64.82x10 ⁴	36.07x10 ⁴	± 3.0	0.65	6.1
180	13400	26800	5360	1650	1.5	2.1	267.1x10 ⁴	170.4x10 ⁴	82.79x10 ⁴	46.07x10 ⁴	± 3.0	0.65	6.0
200	19000	38000	7600	1450			159.5x10 ⁴	126.2x10 ⁴	96.24x10 ⁴	60.2x10 ⁴	± 4.0	0.65	7.8
220	30000	60000	12000	1300			218.8x10 ⁴	174x10 ⁴	128.7x10 ⁴	77.84x10 ⁴	± 4.0	0.70	8.7
240	43000	86000	17200	1200			567.9x10 ⁴	438.3x10 ⁴	301.6x10 ⁴	161.9x10 ⁴	± 4.0	0.70	9.6
260	55000	110000	22000	1000	1.5	2.1	663.8x10 ⁴	539.4x10 ⁴	382.2x10 ⁴	195.5x10 ⁴	± 4.0	0.85	11.3
280	67000	134000	26800	950			773.1x10 ⁴	628.1x10 ⁴	467.9x10 ⁴	266.2x10 ⁴	± 4.0	0.95	12.2

¹⁾ Displacement with n = 1500 rpm

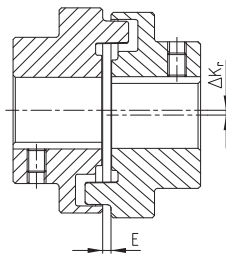
Radial and angular displacements may occur simultaneously. The combined sum of displacements must not exceed the values listed in the table. If requested, coupling is dynamically balanced (semi-key balancing G 6.3 with 1500 rpm). For circumferential speeds exceeding v = 20 m/s dyn. balancing is recommended.

Axial displacement ΔK_a

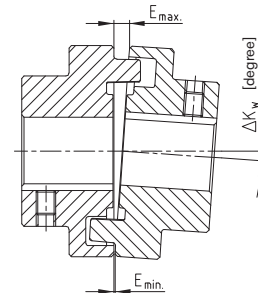


$$L_{\max./\min.} = L + \Delta K_a \text{ [mm]}$$

Radial displacement ΔK_r



Angular displacement ΔK_w



$$\Delta K_w = E_{\max.} - E_{\min.} \text{ [mm]}$$

Advice for assembly

With assembly the coupling halves must be mounted in that coupling and shaft are flush. Alignment must be made in that radial and angular displacement is as small as possible. The service life of coupling and bearings is extended by accurate alignment. Steps must be taken to ensure that the alignment condition does not change during any operating condition. Shaft displacement which cannot be avoided should not exceed the figures specified in the table. Angular and radial displacement may occur simultaneously. The combined sum of displacements must not exceed the values listed in the table above. See KTR assembly instructions, KTR standard 49510 on our homepage www.ktr.com.

General information about the elastomer

Material/hardness	Perbunan [NBR]/78 Shore A
Permanent temperature range [°C]	-30 to +80
Max. temperature (short time) [°C]	-50 to +120
Operating range	General engineering Pump industry ATEX applications Chemical industry Standard applications with average elasticity
Resistant to	Gasoline, diesel Acids, bases Tropics (Salt) water (hot/cold) Oils, greases Propane, butane Natural gas, city gas

Elastomer ring Viton [FKM] 60 Shore A for the high-temperature range on request.

For continuously updated data please refer to our online catalogue at www.ktr.com



Elastomer ring NBR 78 Shore A

ROTEX®

POLY-NORM®

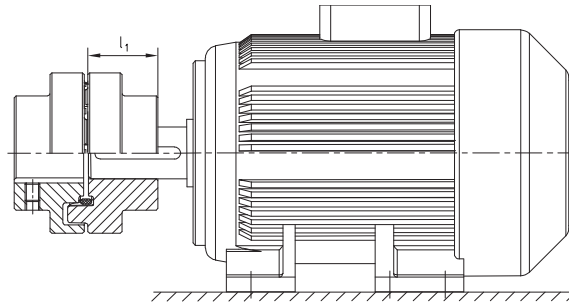
POLY

REVOLUX®

POLY-NORM®

Flexible couplings

Selection of standard IEC motors



POLY-NORM® couplings for standard IEC motors, protection class IP 54/IP 55 (elastomer ring 78 Shore A)															
A. C. motor 50 Hz			Motor power n= 3000 rpm 2 poles		POLY-NORM® coupling size	Motor power n= 1500 rpm 4 poles		POLY-NORM® coupling size	Motor power n= 1000 rpm 6 poles		POLY-NORM® coupling size	Motor power n= 750 rpm 8 poles		POLY-NORM® coupling size	
Size	Shaft end d x l [mm]		Power P [kW]	Torque T [Nm]		Power P [kW]	Torque T [Nm]		Power P [kW]	Torque T [Nm]		Power P [kW]	Torque T [Nm]		
	2 poles	4, 6, 8 poles													
56	9 x 20		0.09	0.32		0.06	0.43		0.037	0.43					
			0.12	0.41		0.09	0.64		0.045	0.52					
63	11 x 23		0.18	0.62		0.12	0.88		0.06	0.7					
			0.25	0.86		0.18	1.3		0.09	1.1					
71	14 x 30		0.37	1.3		0.25	1.8		0.18	2		0.09	1.4		
			0.55	1.9		0.37	2.5		0.25	2.8		0.12	1.8		
80	19 x 40		0.75	2.5	28/32	0.55	3.7	28/32	0.37	3.9	28/32	0.18	2.5	28/32	
			1.1	3.7		0.75	5.1		0.55	5.8		0.25	3.5		
90S	24 x 50		1.5	5		1.1	7.5		0.75	8		0.37	5.3		
90L			2.2	7.4		1.5	10		1.1	12		0.55	7.9		
100L	28 x 60		3	9.8		2.2	15		1.5	15		0.75	11		
						3	20					1.1	16		
112M			4	13		4	27		2.2	22		1.5	21		
132S			5.5	18		5.5	36		3	30		2.2	30		
			7.5	25											
132M	38 x 80				38	7.5	49	38	4	40	38	3	40	38	
									5.5	55					
160M	42 x 110		11	36		11	72		7.5	75	42	4	54		
			15	49	42			42			42	5.5	74	42	
160L			18.5	60		15	98		11	109		7.5	100		
180M	48 x 110		22	71	48	18.5	121	48			48			48	
180L						22	144		15	148		11	145		
200L	55 x 110		30	97		30	196	55	18.5	181	55	15	198	55	
			37	120	55				22	215					
225S	55 x 110	60 x 140				37	240				60	18.5	244		
225M			45	145		45	292	60	30	293		22	290	60	
250M	60 x 140		65 x 140	55	177	60	55	356	65	37	361	65	30	392	65
280S		75 x 140	75	241		75	484		45	438		37	483		
280M			90	289	65	90	581	75	55	535	75	45	587	75	
315S	65 x 140	80 x 170	110	353		110	707		75	727		55	712	85	
315M			132	423		132	849	85	90	873	85	75	971		
			160	513	75	160	1030		110	1070		90	1170	90	
315L			200	641		200	1290	90	132	1280	90	110	1420		
					85				160	1550		132	1710		
315	85 x 170		250	802		250	1600		200	1930	100	160	2070	100	
			315	1010		315	2020	100	250	2410	110	200	2580	110	
			355	1140	90	355	2280		315	3040	125	250	3220		
355	75 x 140	95 x 170	400	1280		400	2570	110	400	3850		315	4060	125	
			500	1600		500	3210								
			560	1790	100	560	3580	125	450	4330	140	355	4570	140	
400	80 x 170	110 x 210	630	2020		630	4030		500	4810		400	5150		
			710	2270	110	710	4540	140	560	5390	160	450	5790	160	
			800	2560		800	5120	160	630	6060		500	6420		
450	90 x 170	120 x 200	900	2880	125	900	5760		710	6830	180	560	7190		
			1000	3200		1000	6400		800	7690		630	8090	180	

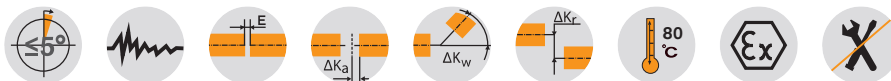
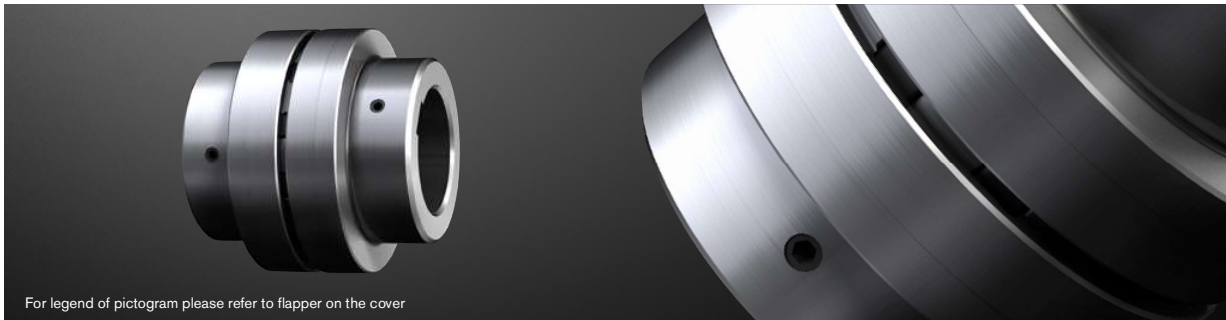
The coupling selection is based on an ambient temperature up to +30 °C. The selection is based on a minimum safety factor of 2 versus the max. coupling torque ($T_{K \max}$). A detailed selection is possible according to catalogue page 15 et seq. Drives with periodical torque curves must be selected according to DIN 740 part 2. If requested, KTR will perform the selection.

Torque T = rated torque according to Siemens catalogue M 11 · 1994/95.

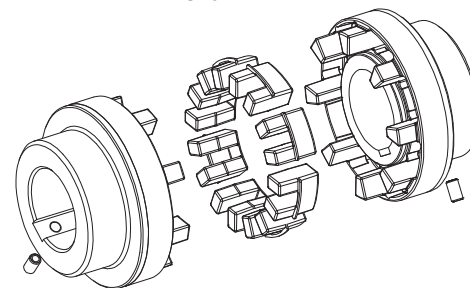
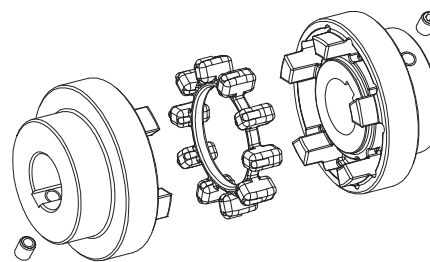
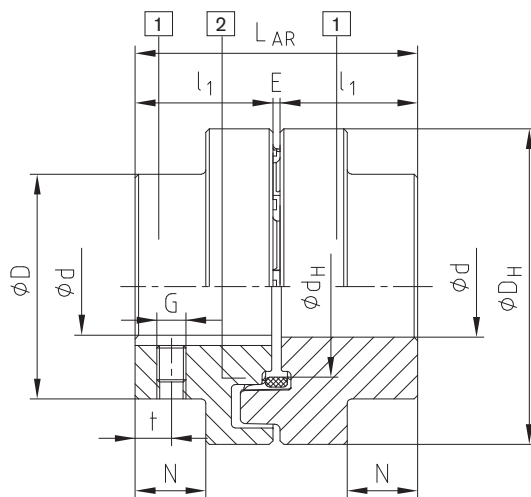
POLY-NORM® AR

Flexible couplings

Two-part



Components



Components of type AR:

1 = Standard hub (GJL)

2 = Elastomer ring (up to size 180: NBR 78 ShA; from size 200: T-PUR® 84 ShA)

POLY-NORM® Type AR														
Size	Elastomer ring ¹⁾ component ²⁾ Torque [Nm]		Max. finish bore d ²⁾	Dimensions [mm]									Mass moment of inertia ³⁾ [kgm²]	Weight [kg]
				General							Setscrew ²⁾			
	LAR	l ₁		E	D _H	D	d _H	N	G	t				
28	40	80.	30	59	28	3	69	46	36.5	12	M5	7	0.0004	0.9
32	60	120	35	68	32	4	78	53	41.5	14	M8	7	0.0008	1.4
38	90	180	40	80	38	4	87	62	50	19.5	M8	10	0.0016	2.0
42	150	300	45	88	42	4	96	69	55.5	20	M8	10	0.0026	2.7
48	220	440	50	101	48	5	106	78	64	24	M8	15	0.0042	3.7
55	300	600	60	115	55	5	118	90	73	29	M8	14	0.0070	5.5
60	410	820	65	125	60	5	129	97	81	33	M8	15	0.0112	6.9
65	550	1100	70	135	65	5	140	105	86	36	M10	20	0.0174	8.8
75	850	1700	80	155	75	5	158	123	100	42.5	M10	20	0.028	13.5
85	1350	2700	90	175	85	5	182	139	116	48.5	M10	25	0.052	19.5
90	2000	4000	95	185	90	5	200	148	128	49	M12	25	0.090	23.2
100	2900	5800	110	206	100	6	224	165	143	55	M12	25	0.160	31.9
110	3900	7800	50-120	226	110	6	250	185	158	60	M16	30	0.317	38.0
125	5500	11000	55-140	256	125	6	280	210	178	70	M16	35	0.570	55.2
140	7200	14400	65-155	286	140	6	315	235	216	76.5	M20	35	1.030	92.6
160	10000	20000	75-175	326	160	6	350	265	246	94.5	M20	45	1.746	126.9
180	13400	26800	75-200	366	180	6	400	300	290	111.5	M20	50	3.239	181.8
200	19000	38000	85-200	408	200	8	450	335	-	126	M24	50	5.728	263.7
220	30000	60000	95-220	448	220	8	500	370	-	140	M24	50	9.489	355.9
240	43000	86000	105-240	488	240	8	550	405	-	154	M24	50	14.963	466.3
260	55000	110000	115-260	530	260	10	650	440	-	158	M24	60	29.504	672.2
280	67000	134000	125-280	570	280	10	700	475	-	172	M24	60	42.451	836.6

¹⁾ Standard material Perbunan [NBR] 78 Shore A, size 140 - 280 double tooth elastomers, for selection see page 14 et seqq.

²⁾ Bores H7 with keyway to DIN 6885 sheet 1 [JS9] and setscrew on the keyway

³⁾ Referring to average bore

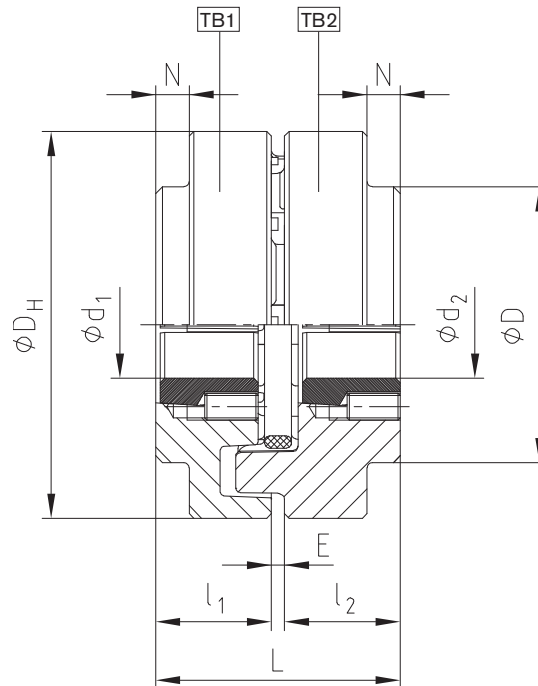
POLY-NORM® AR

Flexible couplings

For taper clamping sleeve



Components



POLY-NORM® for taper clamping sleeve															
Size	Taper clamping sleeve	Dimensions [mm]		Fastening screws ¹⁾ for taper clamping sleeve				Size	Taper clamping sleeve	Dimensions [mm]		Fastening screws ¹⁾ for taper clamping sleeve			
		Max. d1, d2	l1, l2	Size [Inch]	Length [mm]	SW [mm]	T _A [Nm]			Max. d1, d2	l1, l2	Size [Inch]	Length [mm]	SW [mm]	T _A [Nm]
32	1108	25	25.5	1/4"	13	3	5.7	75	2517	60	52.5	1/2"	25	6	49
42	1210	32	31.0	3/8"	16	5	20	85	2517	60	46.5	1/2"	25	6	49
48	1610	40	30.0	3/16"	16	5	20	85	3030	75	82	5/8"	32	8	90
	1615	40	42.5	3/8"	16	5	20	90	3020	75	52.0	5/8"	32	8	92
60	2012	50	38.5	7/16"	22	6	31	100	3535	90	98.0	1/2"	38	10	115
65	2517	60	62.5	1/2"	25	6	49	125	4040	100	111.5	5/8"	45	12	172

¹⁾ Each 2 fastening screws, with 3535/4040 3-off
 For coupling type TB1 screwing on cam side - TB2 screwing on collar side
 Combination possible! Please order our separate dimension sheet (M407045).

Ordering example:	POLY-NORM® 38	AR	Ø38	Ø30
	Coupling size	Type	Finish bore	Finish bore

POLY-NORM® ADR Flexible couplings

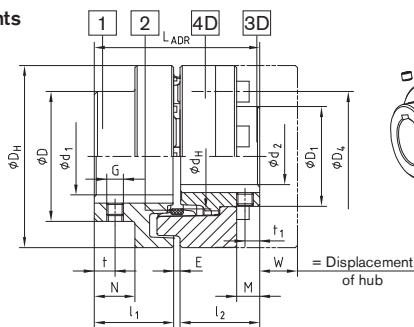
Three-part



For legend of pictogram please refer to flapper on the cover



Components



Size 38 - 125

Size 140 - 280

Components of type ADR (three-part):
1 = Standard hub* (GJL)
2 = Elastomer ring (up to size 180: NBR 78 ShA; from size 200: T-PUR® 84 ShA)
3D = Flange hub (GJS); 4D = Cam ring (GJL)
* To be used preferably on driving side

POLY-NORM® Type ADR

Size	Elastomer ring ¹⁾ (component 2) Torque [Nm]		Dimensions [mm]															
			Max. finish bore ²⁾		General										Setscrew			
	T _{KN}	T _{K max}			d ₁	d ₂	L _{ADR}	l ₁ , l ₂	E	D _H	D	D ₁	d _H	N	M	W	G	t
38	90	180	40	34	80	38	4	87	62	48	50	19.5	11.0	12	M8	10	7	10
42	150	300	45	38	88	42	4	96	69	54	55.5	20	12.0	16	M8	10	7	10
48	220	440	50	44	101	48	5	106	78	62	64	24	13.7	16	M8	15	7	10
55	300	600	60	50	115	55	5	118	90	72	73	29	18.7	15	M8	14	14	10
60	410	820	65	56	125	60	5	129	97	80	81	33	22.2	14	M8	15	15	10
65	550	1100	70	60	135	65	5	140	105	86	86	36	26.7	11	M10	20	20	17
75	850	1700	80	68	155	75	5	158	123	98	100	42.5	27.8	16	M10	20	20	17
85	1350	2700	90	78	175	85	5	182	139	112	116	48.5	33.7	18	M10	25	25	17
90	2000	4000	95	85	185	90	5	200	148	122	128	49	31.5	26	M12	25	25	40
100	2900	5800	110	95	206	100	6	224	165	136	143	55	37.5	28	M12	25	25	40
110	3900	7800	50-120	105	226	110	6	250	185	150	158	60	39.5	30	M16	30	30	80
125	5500	11000	55-140	115	256	125	6	280	210	168	178	70	48.0	35	M16	35	35	80
140	7200	14400	65-155	55-135	286	140	6	315	235	195	216	76.5	47.0	59	M20	35	35	140
160	10000	20000	75-175	65-155	326	160	6	350	265	225	246	94.5	65.0	43	M20	45	45	140
180	13400	26800	75-200	65-175	366	180	6	400	300	255	290	111.5	79.0	33	M20	50	50	140
200	19000	38000	85-200	73-200	408	200	8	450	335	290	320	126	95	7	M24	50	50	240
220	30000	60000	95-220	83-220	448	220	8	500	370	320	354	140	103	8	M24	50	50	240
240	43000	86000	105-240	93-240	488	240	8	550	405	350	388	154	119	1	M24	50	50	240
260	55000	110000	115-260	103-260	530	260	10	650	440	380	445	158	109	34	M24	60	60	240
280	67000	134000	125-280	113-280	570	280	10	700	475	410	478	172	109	29	M24	60	60	240

Selection of cap screws DIN EN ISO 4762 - 12.9

Size	M x l [mm]	z = number	Pitch z x angle	D ₄ [mm]	T _A [Nm] ³⁾	Size	M x l [mm]	z = number	Pitch z x angle	D ₄ [mm]	T _A [Nm] ³⁾
38	M6x16	5	5x72	62	10	110	M16x40	8	8x45	183	210
42	M8x16	5	5x72	69	25	125	M20x40	8	8x45	202	410
48	M8x20	6	6x60	78	25	140	M20x50	8	8x45	237	410
55	M8x20	6	6x60	88	25	160	M20x55	9	9x40	267	410
60	M8x20	6	6x60	98	25	180	M20x60	10	10x36	304	410
65	M10x20	6	6x60	104	49	200	M20x60	10	10x36	342	580
75	M10x25	6	6x60	120	49	220	M24x70	10	10x36	378	1000
85	M12x25	6	6x60	138	86	240	M27x70	10	10x36	416	1500
90	M16x30	6	6x60	149	210	260	M30x90	10	10x36	480	2000
100	M16x30	6	6x60	163	210	280	M30x90	10	10x36	520	2000

¹⁾ Standard material Perbunan [NBR] 78 Shore A, size 140 - 280 double tooth elastomers, for selection see page 14 et seqq.

²⁾ Bores H7 with keyway to DIN 6885 sheet 1 [JS9] and setscrew ³⁾ Screw tightening torques acc. to 8.8

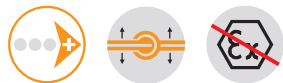
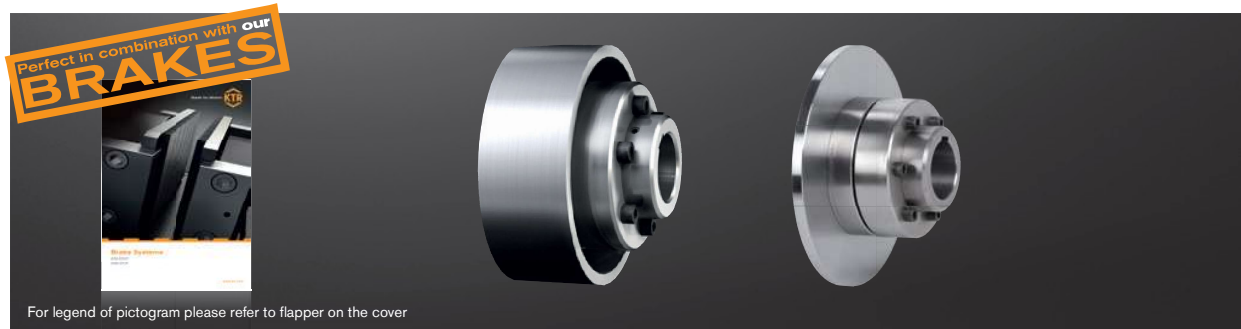
Ordering
example:

POLY-NORM® 65	ADR	d ₁ = Ø55	d ₂ = Ø60
Coupling size	Type	Finish bore	Finish bore

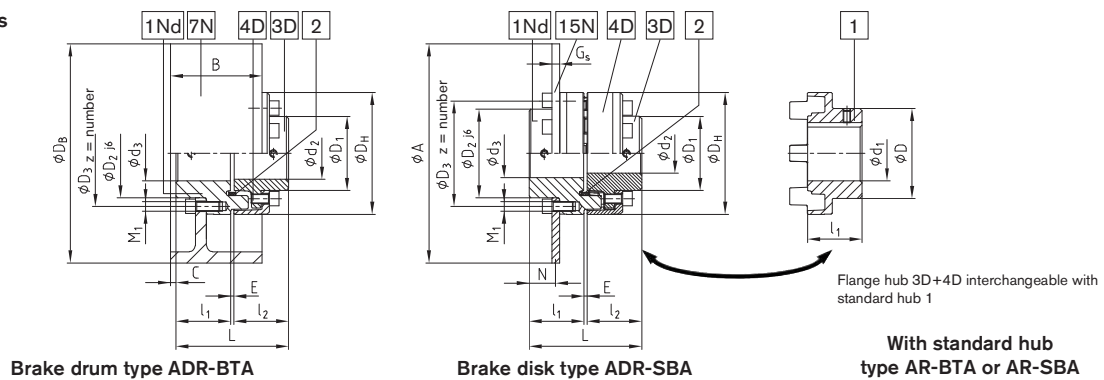
POLY-NORM® BTA and SBA

Flexible couplings

With brake drum/brake disk for brake stop



Components



POLY-NORM® Type AR-BTA, AR-SBA, ADR-BTA and ADR-SBA														
Size	Elastomer ring ¹⁾ (component 2) Torque [Nm]		D, D ₁	Dimensions [mm]										
	TKN	TK max		Max. finish bore			D _H	D ₂	D ₃	z	M ₁	l ₁ , l ₂	E	L
				d ₁	d ₂	d ₃								
38	90	180	For dimensions D, D ₁ please refer to our company catalogue on page 61 and 63	40	34	38	87	61	75	5 x 72°	M6	38	4	80
42	150	300		45	38	42	96	68	82	5 x 72°	M8	42	4	90
48	220	440		50	44	48	106	77	92	6 x 60°	M8	48	5	101
55	300	600		60	50	55	118	88	104	6 x 60°	M8	55	5	115
60	410	820		65	56	60	129	96	114	6 x 60°	M8	60	5	125
65	550	1100		70	60	65	140	104	122	6 x 60°	M10	65	5	135
75	850	1700		80	68	75	158	121	140	6 x 60°	M10	75	5	155
85	1350	2700		90	78	85	182	137	160	6 x 60°	M12	85	5	175
90	2000	4000		95	85	90	200	146	174	6 x 60°	M16	90	5	185
100	2900	5800		110	95	100	224	164	195	6 x 60°	M16	100	6	206
110	3900	7800		50-120	105	50-110	250	184	218	8 x 45°	M16	110	6	226
125	5500	11000		55-140	115	55-125	280	208	245	8 x 45°	M20	125	6	256
140	7200	14400		65-155	55-135	65-140	315	233	276	8 x 45°	M20	140	6	286
160	10000	20000		75-175	65-155	75-160	350	263	308	9 x 40°	M20	160	6	326
180	13400	26800		75-200	65-175	75-180	400	298	349	10 x 36°	M20	180	6	366

POLY-NORM® Type BTA																	POLY-NORM® Type SBA																			
POLY-NORM® size	38	42	48	55	60	65	75	85	90	100	110	125	140	160	180	Max. speed [rpm] with v = 60 m/s ³⁾	POLY-NORM® size	38	42	48	55	60	65	75	85	90	100	110	125	140	160	180	Max. speed [rpm] with v = 60 m/s ³⁾			
ØD _B x _B Brake drum ²⁾	Dimensions C [mm]																ØA _X G Brake disk ²⁾	Dimensions N [mm]																		
160x60	4															7150	200x12.5	13.75														5725				
200x75	9	8	4													5725	250x12.5	13.75	14.75	18.75													4575			
250x95	17	16	20	7	3	0										4575	315x16		13	17	22	26	29	35.5									3625			
315x118		25	21	16	12	9	2.5	-3.5								3625	400x16			17	22	26	29	35.5	41.5	42	48						2850			
400x150			34	28	25	22	15.5	9.5	9	3						2850	500x16				22	26	29	35.5	41.5	42	48	54	64				2275			
500x190											18	12	-2			2275	630x20											46	52	62	69	86	1800			
630x236												20	13	-4		1800	710x20											46	52	62	69	86	104	1600		
710x265													24	7	-11	1600	800x25											43.5	49.5	59.5	66.5	83.5	101.5	1425		
																	900x25														49.5	59.5	66.5	83.5	101.5	1250

¹⁾ Standard material Perbunan [NBR], for selection see page 14 et seq.

²⁾ Steel ³⁾ Dynamic balancing required

Other sizes on request

Ordering example:	POLY-NORM® 38	ADR-BTA	Ø200 x 75	d ₂ = Ø32 NnD	d ₃ = Ø25 NnD
	Coupling size	Type	Brake drum Ø	Component with finish bore	Component with finish bore

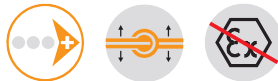
POLY-NORM® ADR-SB

Flexible couplings

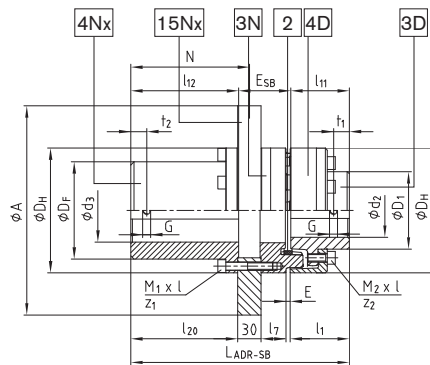
With brake disk for brake stop



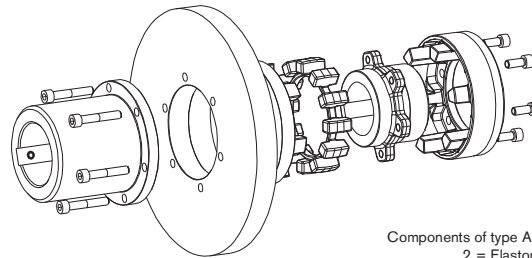
For legend of pictogram please refer to flapper on the cover



Components



Type ADR-SB with brake disk



Components of type ADR-SB:
 2 = Elastomer ring
 3D = Flange hub
 4D = Cam ring
 3N = Driving flange
 15Nx = Brake disk
 4Nx = Coupling flange

POLY-NORM® Type ADR-SB																				
Size	Elastomer ring ¹⁾ (component 2) Torque [Nm]		Max. finish bore		Dimensions [mm]													Setscrew		
	T _{KN}	T _{K max}			d ₂	d ₃	D _H	N	l ₂	E	l ₁₁	l ₁₂	D _F	l ₇	l ₂₀	L _{ADR-SB}	D ₁			
55	300	600	50	65	118	150	55	5	54.7	136.5	88	24.0	135	249.0	72	57.8	M8	15	10	
60	410	820	56	70	129	150	50	5	59.2	136.5	97	25.0	135	255.0	80	59.3	M8	20	10	
65	550	1100	60	75	140	150	65	5	63.7	136.5	105	26.5	135	261.5	86	61.3	M10	20	17	
75	850	1700	68	90	158	150	75	5	74.0	136.5	123	31.5	135	276.5	98	66.0	M10	20	17	
85	1350	2700	78	100	182	150	85	5	84.7	136.5	139	35.0	135	290.0	112	68.8	M10	25	17	
90	2000	4000	85	110	200	150	90	5	89.5	136.5	148	39.5	135	299.5	122	73.5	M12	25	40	
100	2900	5800	95	120	224	190	100	6	95.5	177.0	165	43.0	175	354.0	136	81.5	M12	25	40	
110	3900	7800	105	135	250	190	110	6	105.5	177.0	185	48.0	175	369.0	150	86.5	M16	30	80	
125	5500	11000	115	155	280	195	125	6	120.5	182.0	210	53.0	180	394.0	168	91.5	M16	35	80	
140	7200	14400	135	170	315	195	140	6	130.0	182.0	235	60.5	180	416.5	195	104.5	M20	35	140	
160	10000	20000	155	195	350	195	160	6	150.0	182.0	265	62.5	180	438.5	225	106.5	M20	45	140	

Selection of brake disks and cap screws							
Size	ØA brake disk [mm]/ 30 mm thick ^{2) 3)}	Cap screws DIN EN ISO 4762 for brake disk			Cap screws DIN EN ISO 4762 for flange hub/cam ring		
		M ₁ x l	z ₁ = number	Tightening torque T _A [Nm]	M ₂ x l	z ₂ = number	Tightening torque T _A [Nm]
55	250 - 450	M8x20	6	10	M8x20	6	25
60	250 - 500	M8x20	6	10	M8x20	6	25
65	315 - 500	M8x55	6	35	M10x20	6	49
75	315 - 560	M10x60	6	69	M10x25	6	49
85	355 - 560	M10x60	6	69	M12x25	6	86
90	400 - 710	M12x65	6	120	M16x30	6	210
100	400 - 800	M12x65	6	120	M16x30	6	210
110	450 - 900	M16x75	8	295	M16x40	8	210
125	450 - 900	M16x75	8	295	M20x40	8	410
140	500 - 900	M20x80	8	410	M20x50	8	410
160	560 - 900	M20x90	9	410	M20x55	9	410

¹⁾ Standard material Perbunan [NBR], for selection see page 14 et seqq.

²⁾ Steel

³⁾ For circumferential speeds exceeding v = 20 m/s, dynamic balancing is required (referring to outside diameter ØD_H).

Maximum circumferential speed v = 60 m/s (referring to the diameter of brake disk ØA).

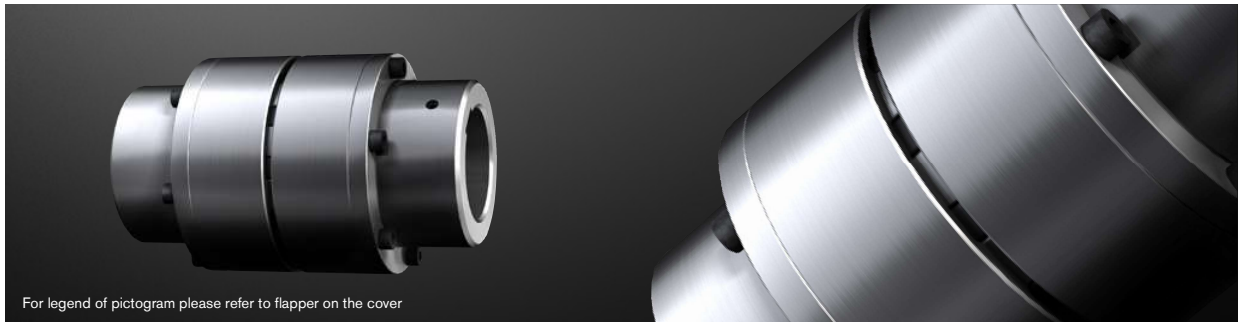
Other sizes on request

Ordering example:	POLY-NORM® 75	ADR-SB	Ø500 x 30	3D d ₂ = Ø60 keyway acc. to DIN	4Nx d ₃ = Ø70 keyway acc. to DIN
	Coupling size	Type	Brake disk ØA/ width	Component with finish bore	Component with finish bore

POLY-NORM® AZR

Flexible couplings

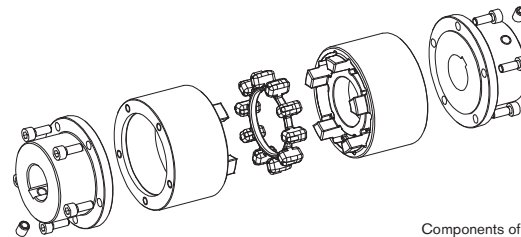
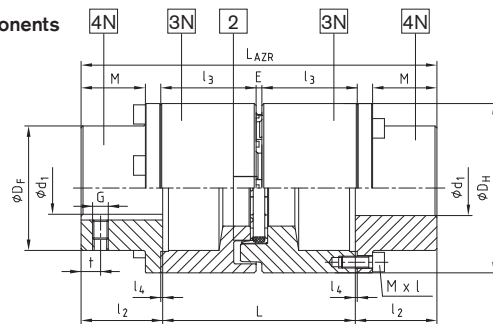
Standard drop-out center design coupling



For legend of pictogram please refer to flapper on the cover



Components



Components of type AZR:
2 = Elastomer ring (NBR 78 ShA)
3N = Driving flange (GJL)
4N = Coupling flange (steel)

POLY-NORM® Type AZR																			
Size	Drop-out center length* L [mm]	Elastomer ring ¹⁾ (component 2) Torque [Nm]		Max. finish bore d ₁ ²⁾	Dimensions [mm]													Mass moment of inertia ³⁾ [kgm ²]	Weight ³⁾ [kg]
					General														
		T _{KN}	T _{K max}		L _{AZR}	l ₂	l ₃	E	l ₄	D _H	D _F	M	Mxl	T _A [Nm]	G	t			
28	100	40	80	34	170	35	49.5	3	1	69	46	26	M6x18	14	M5	7	0.0020	2.4	
32	100	60	120	38	170	35	49	4	1	78	53	26	M6x18	14	M8	7	0.0042	3.2	
	210				69		0.0062										3.9		
38	100	90	180	45	184	42	49	4	1	87	62	33	M6x20	14	M8	10	0.0048	4.3	
	224				69		0.0068										5.1		
42	100	150	300	50	190	45	49	4	1	96	69	35	M6x20	14	M8	10	0.0094	5.1	
	230				69		0.0128										6.0		
48	100	220	440	55	204	52	49	5	1.5	106	78	41.5	M6x20	14	M8	15	0.0170	6.6	
	244				69		0.0216										7.5		
	100				210		49										0.0188	9.4	
55	140	300	600	65	250	55	69	5	1.5	118	88	43.5	M8x25	35	M8	14	0.0240	10.8	
	180				290		89										0.0232	12.2	
	100				220		49										0.0326	11.2	
60	140	410	820	70	260	60	69	5	1.5	129	97	47.5	M8x25	35	M8	15	0.0414	13.0	
	180				300		89										0.0504	14.6	
	100				230		49										0.0564	14.0	
65	140	550	1100	75	270	65	69	5	1.5	140	105	51.5	M8x25	35	M10	20	0.0730	15.8	
	180				310		89										0.0894	17.5	
	140				290		69										0.0824	23.2	
75	180	850	1700	90	330	75	89	5	1.5	158	123	60.5	M10x30	69	M10	20	0.1008	25.6	
	250				400		124										0.1332	29.8	
	140				310		69										0.1570	32.1	
85	180	1350	2700	100	350	85	89	5	1.5	182	139	69.5	M10x30	69	M10	25	0.1658	35.2	
	250				420		124										0.1812	40.7	
	140				320		69										0.2466	38.2	
90	180	2000	4000	110	360	90	89	5	1.5	200	148	73.5	M12x35	120	M12	25	0.2880	42.2	
	250				430		124										0.3566	49.3	
	140				340		69										0.3988	50.0	
100	180	2900	5800	120	380	100	89	6	2	224	165	83	M12x35	120	M12	25	0.4450	54.8	
	250				450		124										0.5465	63.2	

¹⁾ Standard material Perbunan [NBR] 78 Shore A, for selection see page 14 et seqq.

²⁾ Bores H7 with keyway to DIN 6885 sheet 1 [JS9] and setscrew on the keyway

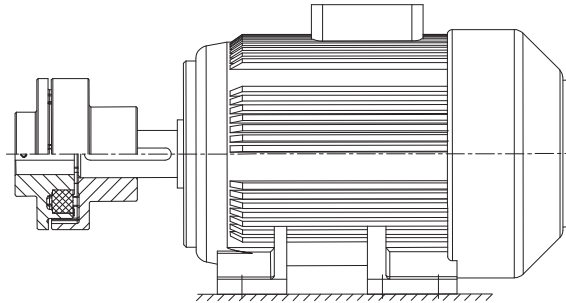
³⁾ Referring to average bore

*For other extendable lengths (L = 120/160/195/215) it is possible to combine two driving flanges 3N with various lengths. (As an example: driving flanges POLY-NORM® 85 for extendable length 140 and 250 produce an extendable length of 195 mm (140 mm + 250 mm = 390 mm; 390 mm/2 = 195 mm))

Ordering example:

POLY-NORM® 42	AZR	140	Ø38	Ø42
Coupling size	Type	Drop-out center length L	Finish bore	Finish bore

Selection of standard IEC motors



POLY couplings for standard IEC standard motors, protection class IP 54/IP 55														
A. C. motor 50 Hz			Motor power n= 3000 rpm 2 poles		POLY coupling size	Motor power n= 1500 rpm 4 poles		POLY coupling size	Motor power n= 1000 rpm 6 poles		POLY coupling size	Motor power n= 750 rpm 8 poles		POLY coupling size
Size	Shaft end d x l [mm]													
	2 poles	4, 6, 8 poles	Power P [kW]	Torque T [Nm]		Power P [kW]	Torque T [Nm]		Power P [kW]	Torque T [Nm]		Power P [kW]	Torque T [Nm]	
56	9 x 20		0.09	0.32		0.06	0.43		0.037	0.43				
			0.12	0.41		0.09	0.64		0.045	0.52				
63	11 x 23		0.18	0.62		0.12	0.88		0.06	0.7				
			0.25	0.86	8	0.18	1.3	8	0.09	1.1	8			8
71	14 x 30		0.37	1.3		0.25	1.8		0.18	2		0.09	1.4	
			0.55	1.9		0.37	2.5		0.25	2.8		0.12	1.8	
80	19 x 40		0.75	2.5		0.55	3.7		0.37	3.9		0.18	2.5	
			1.1	3.7		0.75	5.1		0.55	5.8		0.25	3.5	
90S	24 x 50		1.5	5		1.1	7.5		0.75	8		0.37	5.3	
90L			2.2	7.4		1.5	10		1.1	12		0.55	7.9	
100L	28 x 60		3	9.8	9	2.2	15	9	1.5	15	9	0.75	11	9
						3	20					1.1	16	
112M			4	13		4	27		2.2	22		1.5	21	
132S	38 x 80		5.5	18		5.5	36		3	30		2.2	30	
			7.5	25	10			10			10			10
132M						7.5	49		4	40		3	40	
									5.5	55				
160M	42 x 110		11	36		11	72	12	7.5	75		4	54	
			15	49	12							5.5	74	
160L			18.5	60		15	98	11	109	7.5	100	14		
180M	48 x 110		22	71		18.5	121							
180L						22	144	14				11	145	
									15	148				
200L	55 x 110		30	97		30	196	15	18.5	181	15	15	198	15
			37	120	15				22	215				
225S	55 x 110					37	240	17				18.5	244	17
225M	60 x 140		45	145		45	292		30	293	20	22	290	20
250M	60 x 140	65 x 140	55	177	17	55	356	20	37	361		30	392	
280S	75 x 140		75	241		75	484		45	438	20	37	483	20
280M			90	289	20*	90	581	20	55	535		45	587	
315S			110	353		110	707	22	75	727	22	55	712	22
315M	65 x 140		132	423	20*	132	849		90	873	25	75	971	25
		80 x 170	160	513		160	1030	25	110	1070		90	1170	
315L			200	641		200	1290		132	1280	28	110	1420	28
					22*			28	160	1550		132	1710	
315	85 x 170		250	802		250	1600		200	1930		160	2070	30
			315	1010		315	2020		250	2410	30	200	2580	
			355	1140		355	2280	30						
355	75 x 140	95 x 170	400	1280		400	2570		315	3040		250	3220	35
			500	1600		500	3210		400	3850	35	315	4060	
			560	1790		560	3580	35	450	4330		355	4570	
400	80 x 170	110 x 210	630	2020		630	4030		500	4810		400	5150	40
			710	2270		710	4540		560	5390	40	450	5790	
			800	2560		800	5120	40	630	6060		500	6420	
450	90 x 170	120 x 210	900	2880		900	5760							
			1000	3200		1000	6400							

ROTEX®

POLY-NORM®

POLY

REVOLUX®

Flexible jaw and pin & bush couplings

The coupling selection is based on an ambient temperature up to +30 °C. The coupling was selected for normal operation. The couplings selected have a minimum operating factor $f_{min.} = 1.35$. Drives with periodical torque curves must be selected according to DIN 740 part 2. If requested, KTR will perform the selection.

Torque T = rated torque according to Siemens catalogue M 11 · 1994/95.

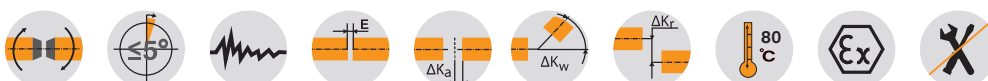
* Dynamic balancing required

POLY PKZ and PKD Flexible couplings

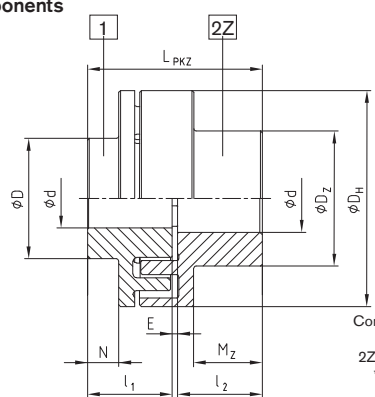
PKZ (two-part) and PKD (three-part)



For legend of pictogram please refer to flapper on the cover

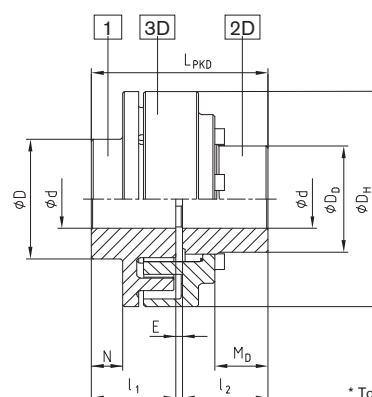


Components



Type PKZ (Z) – (Size 8 to 30)

Components of type PKZ (Z):
1 = Cam section (GJL)
2Z = Pocket element * (GJL)
* To be used preferably on driving side



Type PKD (D) – (Size 15 to 35)

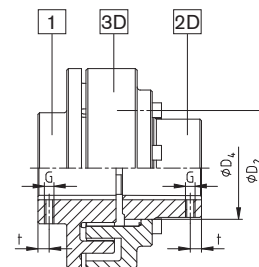
Components of type PKD (D):
1 = Cam section * (GJL)
2D = Flange hub (steel)
3D = Cam ring (GJL)
* To be used preferably on driving side

POLY Type PKZ and PKD																	
Size	Rated torque ¹⁾ T _{KN} [Nm]	Max. speed ²⁾ n [rpm]	Max. finish bore d			Dimensions [mm]											
			Component 1	Component 2Z	Component 2D	D _H	D	D _Z	D _D	l ₁ , l ₂	M _Z	M _D	N	E	D ₂	D ₄ (H7/h7)	L _{PKZ} /L _{PKD}
8 (Z)	42	5000	20	28	—	86	43	50	—	35	25	—	3	3	—	—	73
9 (Z)	72	5000	28	38	—	97	55	65	—	41	30	—	7	3	—	—	85
10 (Z)	100	5000	32	42	—	107	60	70	—	45	35	—	10	4	—	—	94
12 (Z)	170	5000	38	48	—	131	70	80	—	55	43	—	12	4	—	—	114
14 (Z)	210	4800	45	55	—	142	80	93	—	60	46	—	17	4	—	—	124
15 (Z,D)	320	4300	50	60	50	157	90	100	74.5	65	52	33	21	4	90	75	134
17 (Z,D)	400	3800	60	65	60	176	100	110	87	70	56	43.5	26	4	106	90	144
20 (Z,D)	820	3300	65	75	70	205	115	127	104	80	65	45	23	4	123	105	164
22 (Z)	1100	3000	85	85	—	224	140	140	—	90	75	—	38	4	—	—	184
25 (Z,D)	1600	2700	90	90	95	257	150	150	138	100	84	67	43	5	162	140	205
28 (Z)	2500	2350	100	100	110	288	165	165	158	110	90	65	44	5	178	160	225
30 (Z,D)	3950	2200	110	110	110	308	180	180	165	130	108	89	58	5	202	170	265
35 (D)	6100	1850	130	—	145	373	210	—	209	160	—	102	70	5	240	210	325

¹⁾ Maximum torque T_{K max} = T_{KN} x 2; elastomer: standard material Perbunan [NBR] 92 Shore A; hub: standard material GJL

²⁾ Speeds for v = 30 m/s. For circumferential speeds exceeding v = 30 m/s, we recommend dynamic balancing

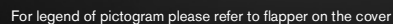
³⁾ Referring to average bore



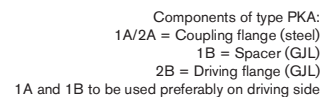
Ordering example:	POLY	PKD	28	d ₁ = Ø90	d ₂ = Ø80
	Coupling type	Type	Size	Finish bore component 1	Finish bore component 2

Flexible jaw and pin & bush couplings

ROTEX®



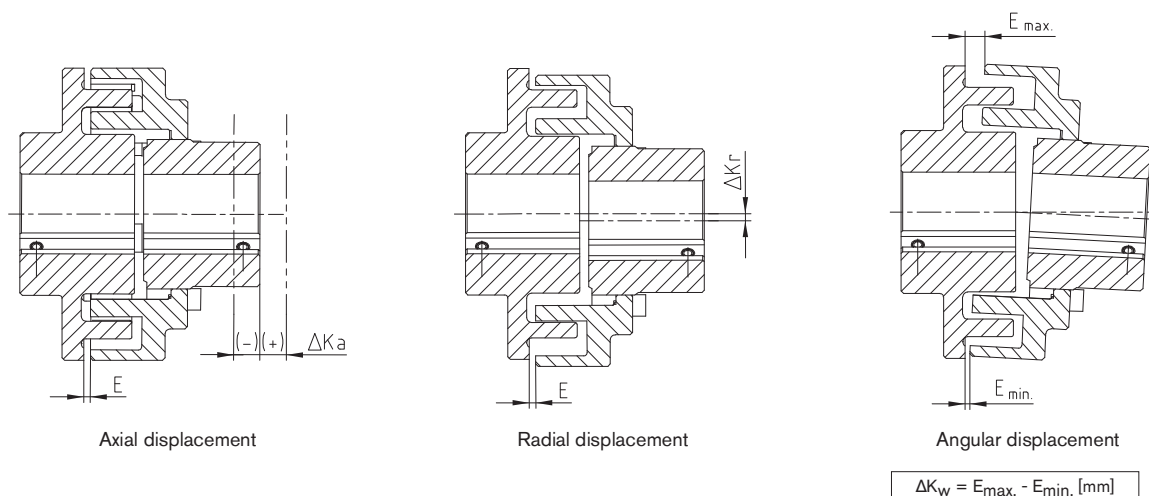
POLY-NORM®



POLY

REVOLX®

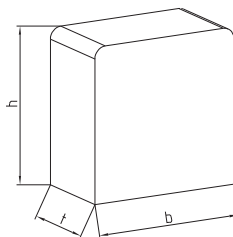
Displacements / elastomer sets / screws



Radial and angular displacements may occur simultaneously.

The combined sum $V = \Delta K_r \text{ [mm]} + (E_{\max.} \text{ [mm]} - E_{\min.} \text{ [mm]})$ should not exceed the values listed in the table.

Displacements [mm]														
Coupling size	8	9	10	12	14	15	17	19	20	22	25	28	30	35
Max. axial displacement ΔK_a [mm]	± 1	± 1	± 1	± 2	± 2	± 2	± 2	± 2	± 2	± 2	± 2	± 2	± 2	± 3
Max. radial displacement ΔK_r or max. angular displacement ΔK_w or sum V	n=750 rpm	0.8	0.8	0.8	0.8	0.8	1.0	1.0	1.0	1.0	1.0	1.0	1.2	1.2
n=1000 rpm	0.7	0.7	0.7	0.7	0.7	0.9	0.9	0.9	0.9	0.9	0.9	0.9	1.1	1.1
n=1500 rpm	0.5	0.5	0.5	0.5	0.5	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.9



Elastomer sets NBR (cuboid)														
Coupling size	8	9	10	12	14	15	17	19	20	22	25	28	30	35
Size of set	1			2			3			3a	4	3b	4Ü	5
Number of sets	8	10	10	10	10	12	12	12	12	16	16	16	16	20
Dimensions of elastomer set w x d x h [mm]	b	18.4			24.9			27.2			27.7	34.9	29.6	34.8
	t	10			15.3			16.1			18.4	19.6	18.4	20.1
	h	18.9			23.9			24.6			26.8	34.6	29.6	35

Type PKD - Dimensions of cap screws acc. to DIN EN ISO 4762														
Coupling size	8	9	10	12	14	15	17	19	20	22	25	28	30	35
Screw size	M	—	—	—	—	M8	M8	M8	M10	M8	M10	M10	M12	M12
	I	—	—	—	—	30	25	25	30	30	30	40	40	55
Number	—	—	—	—	—	6	6	6	6	8	8	8	8	10
Tightening torque T_A [Nm]	—	—	—	—	—	25	25	25	49	25	49	49	86	86
Type PKA - Dimensions of cap screws acc. to DIN EN ISO 4762														
Screw size	M	M6	M6	M6	M8	M8	M10	M10	—	M10	—	—	—	—
	I	16	18	18	20	20	25	25	—	30	—	30	—	—
Number	4	5	5	5	5	6	6	—	6	—	8	—	—	—
Tightening torque T_A [Nm]	10	10	10	25	25	49	49	—	49	—	49	—	—	—

REVOLEX®

Flexible pin & bush coupling

Flexible jaw and
pin & bush couplings

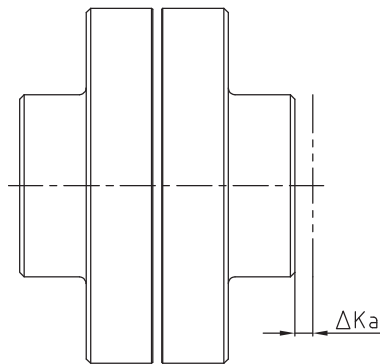
Technical data

REVOLEX® KX-D Technical data											
Size	Torque [Nm] NBR 80 ShA			Cast iron		Steel		Torsion spring stiffness C dyn. [Nm/rad]			
	Rated TKN	Max. TK max	Vibratory TKW	Max. speed [rpm] with v = 35 m/s	Max. bore [mm]	Max. speed [rpm] with v = 60 m/s	Max. bore [mm]	0.25 T _{KN}	0.50 T _{KN}	0.75 T _{KN}	1.00 T _{KN}
KX-D 75	4300	8600	1720	–	–	4500	100	6.73x10 ⁵	1.04x10 ⁶	1.54x10 ⁶	2.17x10 ⁶
KX-D 85	5500	11000	2200	–	–	4175	110	8.66x10 ⁵	1.32x10 ⁶	1.95x10 ⁶	2.73x10 ⁶
KX-D 95	7200	14400	2880	–	–	3845	125	1.11x10 ⁶	1.70x10 ⁶	2.49x10 ⁶	3.47x10 ⁶
KX-D 105	9400	18800	3760	2000	110	3475	130	1.45x10 ⁶	2.21x10 ⁶	3.23x10 ⁶	4.51x10 ⁶
KX-D 120	15200	30400	6080	1800	125	3100	150	1.78x10 ⁶	2.48x10 ⁶	3.54x10 ⁶	4.85x10 ⁶
KX-D 135	20000	40000	8000	1600	140	2725	170	2.34x10 ⁶	3.27x10 ⁶	4.65x10 ⁶	6.36x10 ⁶
KX-D 150	25000	50000	10000	1450	160	2500	190	2.94x10 ⁶	4.12x10 ⁶	5.87x10 ⁶	8.06x10 ⁶
KX-D 170	41000	82000	16400	1250	180	2150	220	4.36x10 ⁶	7.91x10 ⁶	1.14x10 ⁷	1.50x10 ⁷
KX-D 190	54000	108000	21600	1100	205	1900	245	5.78x10 ⁶	1.05x10 ⁷	1.52x10 ⁷	1.98x10 ⁷
KX-D 215	67500	135000	27000	1000	230	1725	275	7.21x10 ⁶	1.30x10 ⁷	1.89x10 ⁷	2.47x10 ⁷
KX-D 240	98000	196000	39200	900	250	1550	310	1.19x10 ⁷	2.52x10 ⁷	3.84x10 ⁷	5.17x10 ⁷
KX-D 265	134000	268000	53600	800	285	1375	350	1.62x10 ⁷	3.44x10 ⁷	5.26x10 ⁷	7.07x10 ⁷
KX-D 280	170000	340000	68000	720	315	1225	385	2.05x10 ⁷	4.36x10 ⁷	6.66x10 ⁷	8.97x10 ⁷
KX-D 305	205000	410000	82000	675	330	1150	405	2.49x10 ⁷	5.27x10 ⁷	8.05x10 ⁷	1.08x10 ⁸
KX-D 330	265000	530000	106000	625	355	1075	435	3.20x10 ⁷	6.78x10 ⁷	1.04x10 ⁸	1.40x10 ⁸
KX-D 355	350000	700000	140000	575	380	975	450	4.26x10 ⁷	8.99x10 ⁷	1.37x10 ⁸	1.85x10 ⁸
KX-D 370	430000	860000	172000	535	450	900	530	4.92x10 ⁷	1.04x10 ⁸	1.59x10 ⁸	2.14x10 ⁸
KX-D 470	520000	1040000	208000	–	–	855	520	6.25x10 ⁷	1.33x10 ⁸	2.03x10 ⁸	2.74x10 ⁸
KX-D 520	810000	1620000	324000	–	–	740	According to customer specification	9.83x10 ⁷	2.08x10 ⁸	3.18x10 ⁸	4.28x10 ⁸
KX-D 590	1000000	2000000	400000	–	–	660		1.21x10 ⁸	2.56x10 ⁸	3.91x10 ⁸	5.26x10 ⁸
KX-D 650	1350000	2700000	540000	–	–	590		1.63x10 ⁸	3.47x10 ⁸	5.30x10 ⁸	7.14x10 ⁸

■ = Years of experience with applications at customer sites and additional test series in the KTR test field in Rheine enabled us to determine potentials allowing for an increase of the rated and maximum torques with some sizes of this series.

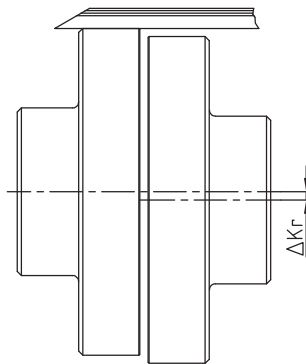
If requested, coupling is dynamically balanced (semi-key balancing G 6.3 with speed on request of customer). For circumferential speeds exceeding v = 30 m/s dyn. balancing is recommended.

Axial displacement ΔK_a

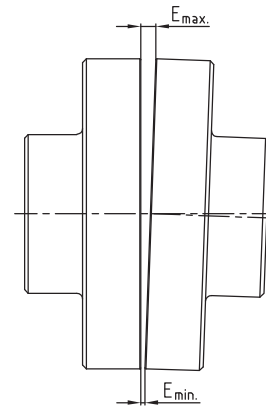


$$L_{\max/\min} = L + \Delta K_a \text{ [mm]}$$

Radial displacement ΔK_r



Angular displacement ΔK_w



$$\Delta K_w = E_{\max} - E_{\min} \text{ [mm]}$$

Displacements																						
Size (KX and KX-D)		75	85	95	105	120	135	150	170	190	215	240	265	280	305	330	355	370	470	520	590	650
Max. axial displacement ΔK_A [mm]		±1.5	±1.5	±1.5	±2	±2	±2	±2	±2.5	±2.5	±2.5	±2.5	±2.5	±2.5	±2.5	±4	±4	±4	±4	±4	±4	±4
Max. radial displacement ΔK_r [mm] or max. angular displacement ΔK_W [mm] with speed n	250 rpm	0.95	1.1	1.1	1.2	1.3	1.4	1.5	1.7	1.9	2.0	2.2	2.5	2.7	2.9	3.1	3.3	3.5	3.8	4.4	4.9	5.4
	500 rpm	0.70	0.80	0.80	0.9	0.9	1.0	1.1	1.2	1.3	1.4	1.6	1.7	1.9	2.0	2.2	2.3	2.5	2.8	3.1	3.5	3.8
	750 rpm	0.60	0.65	0.65	0.7	0.8	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.6	1.7	1.8	1.9	2.0	2.2	2.4	—	—
	1000 rpm	0.50	0.55	0.55	0.6	0.7	0.7	0.8	0.9	0.9	1.0	1.1	1.2	1.4	1.4	1.5	1.7	1.8	—	—	—	—
	1500 rpm	0.40	0.45	0.45	0.5	0.5	0.6	0.6	0.7	0.8	0.8	0.9	1.0	—	—	—	—	—	—	—	—	—
	2000 rpm	0.35	0.40	0.40	0.4	0.5	0.5	0.5	0.6	0.7	—	—	—	—	—	—	—	—	—	—	—	—
	3000 rpm	0.30	0.35	0.35	0.4	0.4	0.4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Advice for assembly

The permissible displacement figures of the flexible REVOLEX® KX couplings specified are general standard values taking into account the load of the coupling up to the rated torque T_{KN} of the coupling and an ambient temperature of +30 °C. The displacement figures may only be used one by one, if they appear simultaneously, they must be limited in proportion. Care should be taken to maintain the distance dimension E accurately in order to allow for axial clearance of the coupling while in operation. See KTR assembly instructions, KTR standard 49410 on our homepage www.ktr.com.

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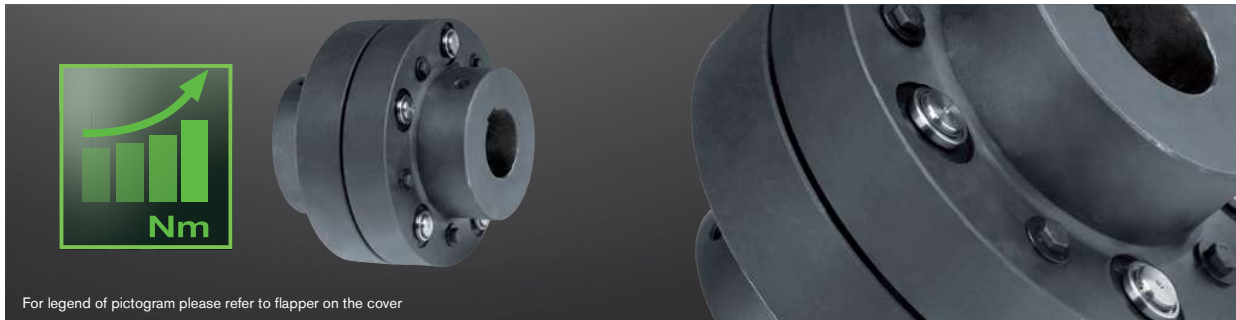
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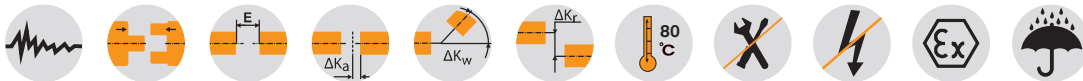
REVOLEX® KX-D

Flexible pin & bush coupling

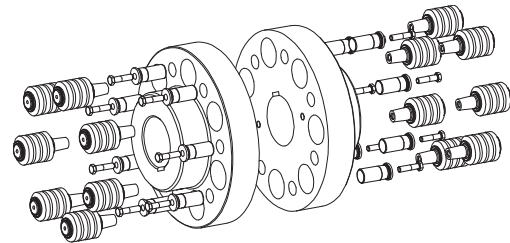
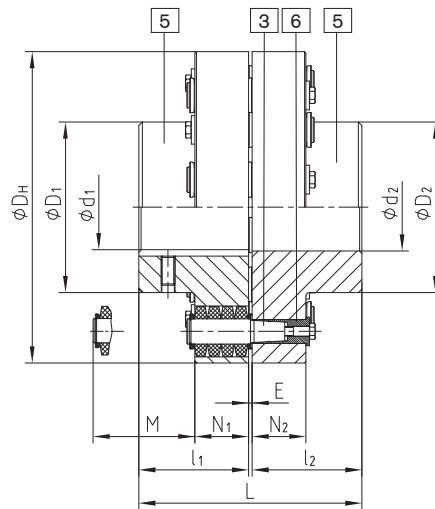
Material cast



For legend of pictogram please refer to flapper on the cover



Components



Components of type KX-D:

5 = Hub part 5

3 = Pins complete

6 = KX-D sleeve (hardened and corrosion-resistant)

REVOLEX® Type KX-D													
Size	Torque ¹⁾ [Nm]		Max. speed ²⁾ [rpm]	Finish bore (min. - max.)	Dimensions [mm]							Mass moment of inertia ³⁾ [kgm²]	Weight ³⁾ [kg]
	T _{KN}	T _{Kmax.}			L	I ₁ , I ₂	E	D _H	D ₁ , D ₂	N ₁ , N ₂	M*		
KX-D 105	9400	18800	2000	38-110	237	117	3	330	180	56	76	0.907	68
KX-D 120	15200	30400	1800	45-125	270	132	6	370	206	76	100	1.867	108
KX-D 135	20000	40000	1600	75-140	300	147	6	419	230	76	100	3.144	145
KX-D 150	25000	50000	1450	85-160	336	165	6	457	256	76	100	4.573	180
KX-D 170	41000	82000	1250	95-180	382	188	6	533	292	92	130	10.259	291
KX-D 190	54000	108000	1100	110-205	428	211	6	597	330	92	130	16.601	385
KX-D 215	67500	135000	1000	125-230	480	237	6	660	368	92	130	25.495	498
KX-D 240	98000	196000	900	140-250	534	264	6	737	407	122	170	50.147	760
KX-D 265	134000	268000	800	160-285	590	292	6	826	457	122	170	80.796	997
KX-D 280	170000	340000	720	180-315	628	311	6	927	508	122	170	129.979	1301
KX-D 305	205000	410000	675	180-330	654	324	6	991	533	122	170	170.016	1509
KX-D 330	265000	530000	625	200-355	666	330	6	1067	572	122	170	227.451	1755
KX-D 355	350000	700000	575	225-380	721	356	9	1156	610	164	220	415.259	2263
KX-D 370	430000	860000	535	225-450	773	382	9	1250	720	164	220	586.686	2701

■ = Years of experience with applications at customer sites and additional test series in the KTR test field in Rheine enabled us to determine potentials allowing for an increase of the rated and maximum torques with some sizes of this series.

* Drop-out center dimension required

¹⁾ Standard material Perbunan [NBR] 80 Shore A, for selection see page 18 et seqq.

²⁾ Higher speeds on request.

³⁾ Referring to max. bore

Finish bore according to ISO fit H7, feather keyway according to DIN 6885, sheet 1 [JS9].

If requested, coupling is dynamically balanced (semi-key balancing G 6.3 with speed on request of customer). For circumferential speeds exceeding $v = 30$ m/s dyn. balancing is recommended.

■ = Pilot bored available from stock

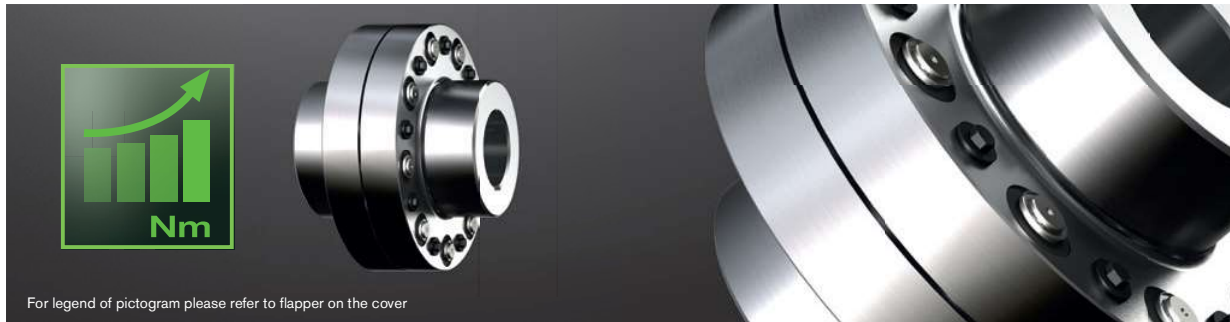
Ordering example:

REVOLEX® KX-D 170	GJL	Ø120	Ø150
Type and size of coupling	Material	Finish bore	Finish bore

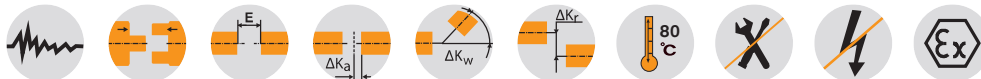
REVOLEX® KX-D

Flexible pin & bush coupling

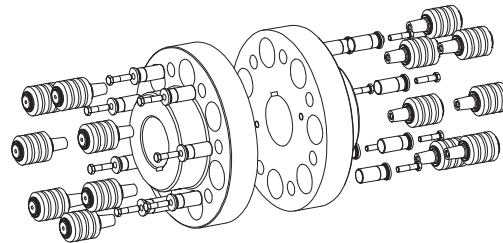
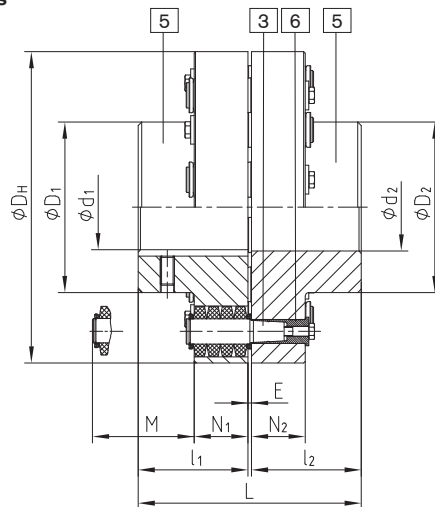
Material steel



For legend of pictogram please refer to flapper on the cover



Components



Components of type KX-D:

5 = Hub part 5

3 = Pins complete

6 = KX-D sleeve (hardened and corrosion-resistant)

REVOLEX® Type KX-D													
Size	Torque ¹⁾ [Nm]		Max. speed ²⁾ [rpm]	Finish bore (min. - max.)	Dimensions [mm]							Mass moment of inertia ³⁾ [kgm²]	Weight ³⁾ [kg]
	T _{KN}	T _{Kmax.}			d ₁ , d ₂	L	l ₁ , l ₂	E	D _H	D ₁ , D ₂	N ₁ , N ₂		
KX-D 75	4300	8600	4500	0-100	193	95	3	255	136	56	76	0.325	39
KX-D 85	5500	11000	4175	0-110	213	105	3	274	152	56	76	0.440	46
KX-D 95	7200	14400	3825	0-125	227	112	3	298	168	56	76	0.624	56
KX-D 105	9400	18800	3475	0-130	237	117	3	330	180	56	76	0.907	80
KX-D 120	15200	30400	3100	0-150	270	132	6	370	206	76	100	1.867	124
KX-D 135	20000	40000	2725	75-170	300	147	6	419	230	76	100	3.144	165
KX-D 150	25000	50000	2500	85-190	336	165	6	457	256	76	100	4.573	205
KX-D 170	41000	82000	2150	95-220	382	188	6	533	292	92	130	10.259	322
KX-D 190	54000	108000	1900	110-245	428	211	6	597	330	92	130	16.601	431
KX-D 215	67500	135000	1725	125-275	480	237	6	660	368	92	130	25.495	559
KX-D 240	98000	196000	1550	140-310	534	264	6	737	407	122	170	50.147	833
KX-D 265	134000	268000	1375	160-350	590	292	6	826	457	122	170	80.796	1099
KX-D 280	170000	340000	1225	180-385	628	311	6	927	508	122	170	129.979	1436
KX-D 305	205000	410000	1150	180-405	654	324	6	991	533	122	170	170.016	1669
KX-D 330	265000	530000	1075	200-435	666	330	6	1067	572	122	170	227.451	1954
KX-D 355	350000	700000	975	225-450	721	356	9	1156	610	164	220	415.259	2451
KX-D 370	430000	860000	900	225-530	773	382	9	1250	720	164	220	584.686	2925
KX-D 470	520000	1040000	855	240-520 ⁴⁾	969 ⁴⁾	480 ⁴⁾	9	1340	705 ⁴⁾	164	220	785.489	3631
KX-D 520	810000	1620000	760	240-520 ⁴⁾	1089 ⁴⁾	540 ⁴⁾	9	1540	780 ⁴⁾	164	220	1264.725	5155
KX-D 590	1000000	2000000	680	260-590 ⁴⁾	1212 ⁴⁾	600 ⁴⁾	12	1735	885 ⁴⁾	164	220	2081.885	6895
KX-D 650	1350000	2700000	610	280-650 ⁴⁾	1332 ⁴⁾	660 ⁴⁾	12	1935	975 ⁴⁾	164	220	3228.297	8893

¹⁾ = Years of experience with applications at customer sites and additional test series in the KTR test field in Rheine enabled us to determine potentials allowing for an increase of the rated and maximum torques with some sizes of this series.

* Drop-out center dimension required ¹⁾ Standard material Perbunan [NBR] 80 Shore A, for selection see page 18 et seqq.

²⁾ Higher speeds on request. ³⁾ Referring to max. bore ⁴⁾ Variable according to customer's requests

Finish bore according to ISO fit H7, feather keyway according to DIN 6885, sheet 1 [JS9]. If requested, coupling is dynamically balanced (semi-key balancing G 6.3 with speed on request of customer). For circumferential speeds exceeding $v = 30$ m/s dyn. balancing is recommended.

■ = Pilot bored available from stock

Ordering example:	REVOLEX® KX-D 170	Steel	Ø120	Ø150
	Type and size of coupling	Material	Finish bore	Finish bore

REVOLEX® KX-D SB

Flexible pin & bush coupling

With brake disk



REVOLEX® KX-D Type SB													
Size	Torque ¹⁾ [Nm] KX-D		Finish bore KX-D (min. - max.)		Dimensions [mm]								
	TKN	TK max	GJL d ₁ , d ₂	Steel d ₁ , d ₂	L	I ₁ , I ₂	E	D _H	D ₁	N ₁	N ₂	N ₃	M*
105	9400	18800	34-110	0-130	237	117	3	330	180	56	29	55	76
120	15200	30400	50-125	0-150	270	132	6	370	206	76	45	75	100
135	20000	40000	70-140	70-170	300	147	6	419	230	76	45	75	100
150	25000	50000	82-160	82-190	336	165	6	457	256	76	45	75	100
170	41000	82000	95-180	95-220	382	188	6	533	292	92	62	91	130
190	54000	108000	110-205	110-245	428	211	6	597	330	92	62	91	130
215	67500	135000	125-230	125-275	480	237	6	660	368	92	62	91	145
240	98000	196000	140-250	140-310	534	264	6	737	407	122	75	121	167

■ = Years of experience with applications at customer sites and additional test series in the KTR test field in Rheine enabled us to determine potentials allowing for an increase of the rated and maximum torques with some sizes of this series.

Selection of coupling/brake disk dimension N						
Size	Brake disk ØA x b ²⁾					
	Ø560x30	Ø630x30	Ø710x30	Ø800x30	Ø900x30	Ø1000x30
	KX-D	KX-D	KX-D	KX-D	KX-D	KX-D
105	47	47				
120	42	42				
135		57	57			
150			75	75		
170			82	82		
190				105	105	
215				131	131	131
240				128	128	128

* Drop-out center dimension required

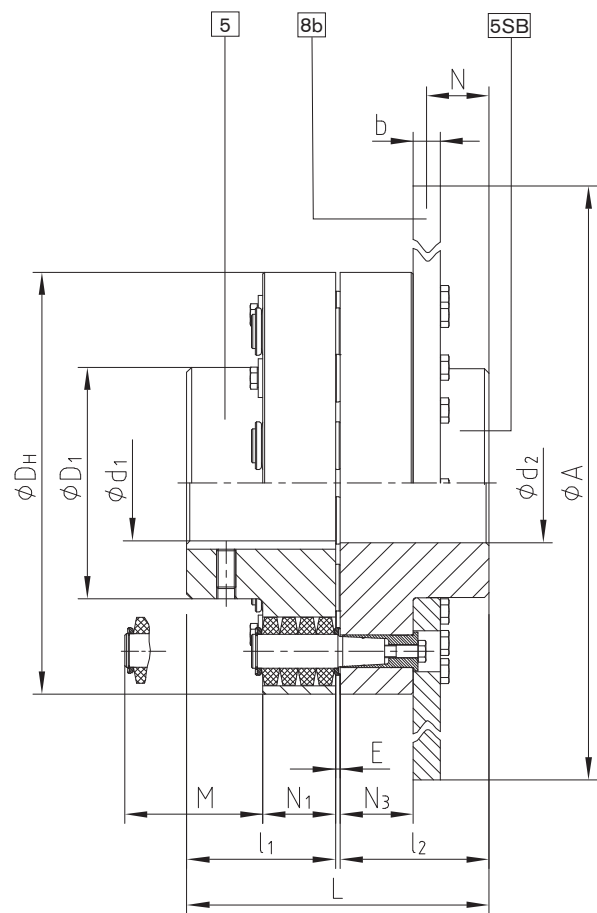
¹⁾ Standard material Perbunan [NBR] 80 Shore A, for selection see page 18 et seqq.

²⁾ Maximum circumferential speed v = 60 m/s referring to the maximum outside diameter.

Finish bore according to ISO fit H7, feather keyway according to DIN 6885, sheet 1 [JS9].

If requested, coupling is dynamically balanced (semi-key balancing G 6.3 with speed on request of customer). For circumferential speeds exceeding v = 30 m/s dynamic balancing is necessary (referring to outside diameter ØA).

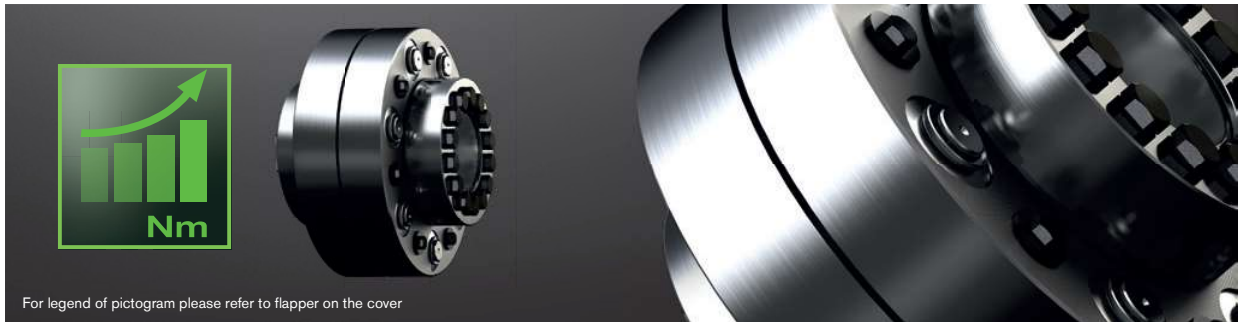
Ordering example:	REVOLEX® KX 170	SB	Ø710 x 30	1 - Ø120	2SB - Ø150
	Type and size of coupling	Type	Brake disk	Finish bore	Finish bore



REVOLEX® KX-D with KTR 650 clamping set

Flexible pin & bush coupling

Material steel



REVOLEX® KX-D						CLAMPEX® KTR 650						
Size	D _H	L ₁ , L ₂	N	E	T _{KN} ¹⁾ [Nm]	Dimensions [mm]		Screws				Transmittable torque T [Nm]
						Max. perm. size d x D	T _K	Size	Length	Number	T _A [Nm]	
105	330	123	56	3	9400	100 x 197	156	M16	75	14	250	18800
120	370	149	76	6	15200	110 x 215	166	M20	90	10	490	22400
135	419	157	76	6	20000	120 x 230	186	M20	90	14	490	35200
150	457	168	76	6	25000	140 x 290	216	M20	100	16	490	46700
170	533	205	92	6	41000	180 x 340	276	M24	130	16	840	85800
190	597	214	92	6	54000	220 x 405	320	M27	140	18	1250	148600
215	660	232	92	6	67500	260 x 460	356	M27	160	21	1250	192900
240	737	254	122	6	98000	240 x 430	340	M27	150	20	1250	175400
265	826	280	122	6	134000	320 x 550	402	M27	180	24	1250	248900
280	927	313	122	6	170000	390 x 630	486	M30	200	24	1700	368300
305	991	321	122	6	205000	440 x 700	534	M30	220	28	1700	472100
330	1067	321	122	6	265000	440 x 700	534	M30	220	28	1700	472100

■ = Years of experience with applications at customer sites and additional test series in the KTR test field in Rheine enabled us to determine potentials allowing for an increase of the rated and maximum torques with some sizes of this series.

¹⁾ Standard material Perbunan [NBR] 80 Shore A, for selection see page 18 et seqq.

If requested, coupling is dynamically balanced (semi-key balancing G 6.3 with speed on request of customer). For circumferential speeds exceeding v = 30 m/s dynamic balancing is necessary (referring to outside diameter ØA).

CLAMPEX® KTR 650							
d x D	Dimensions [mm]		Screws				Transmittable torque T [Nm]
	B1/B2	T _K	Size	Length	Number	T _A [Nm]	
100 x 215	73	166	M20	90	10	490	22400
110 x 230	81	186	M20	90	14	490	33600
120 x 290	92	216	M20	100	16	490	42100
130 x 290	92	216	M20	100	16	490	46700
140 x 320	102	234	M24	110	14	840	63600
150 x 320	102	234	M24	110	14	840	63600
160 x 340	113	276	M24	130	16	840	85800
170 x 340	113	276	M24	130	16	840	85800
180 x 370	117	290	M27	140	16	1,250	119700
190 x 370	117	290	M27	140	16	1,250	119700
200 x 405	122	320	M27	140	18	1,250	148600
210 x 405	122	320	M27	140	18	1,250	148600
220 x 430	132	340	M27	150	20	1,250	175400
230 x 430	132	340	M27	150	20	1,250	175400
240 x 460	140	356	M27	160	21	1,250	192900
250 x 460	140	356	M27	160	21	1,250	192900
260 x 485	147	360	M27	180	21	1,250	195000
270 x 485	147	360	M27	180	21	1,250	195000
280 x 520	150	380	M27	180	21	1,250	205900
290 x 520	150	380	M27	180	21	1,250	205900
300 x 550	158	402	M27	180	24	1,250	248900
310 x 550	158	402	M27	180	24	1,250	248900
320 x 570	164	424	M27	180	24	1,250	262500
330 x 570	164	424	M27	180	24	1,250	262500
340 x 610	171	454	M30	190	24	1,700	344000
350 x 610	171	454	M30	190	24	1,700	344000

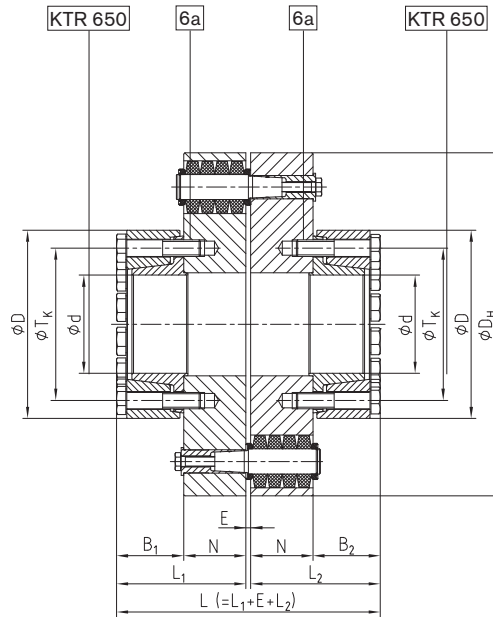
Ordering example:

REVOLEX® KX-D 170	Steel	KTR 650 Ø120	KTR 650 Ø150
Type and size of coupling	Material	KTR 650 for shaft diameter	KTR 650 for shaft diameter

ROTEX®

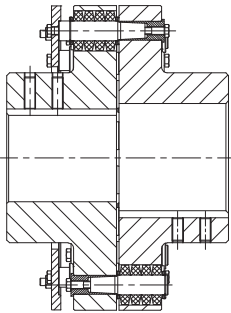
POLY-NORM®

Components

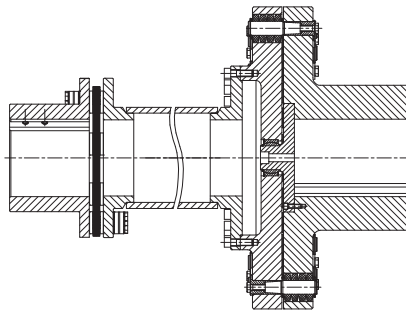


Other types

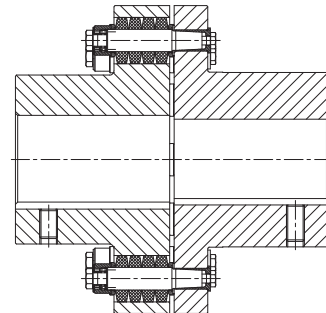
Type AB with limitation of axial backlash



Intermediate shaft type with RADEX®-N

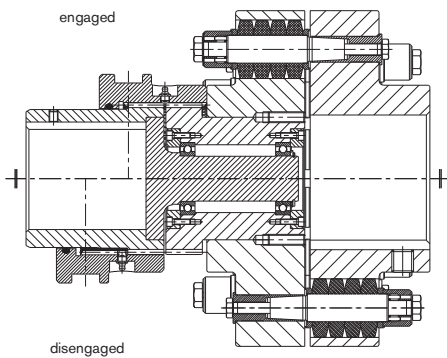


Backlash-free type

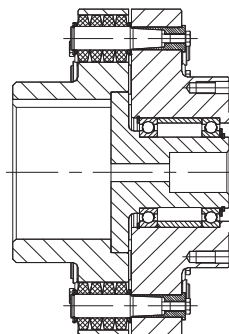


POLY

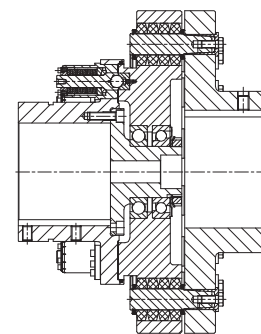
Type KX-D SD
shiftable with shiftable linkage



Type KX-D
with cardan shaft connection



Type KX-D
with KTR-SI FRE

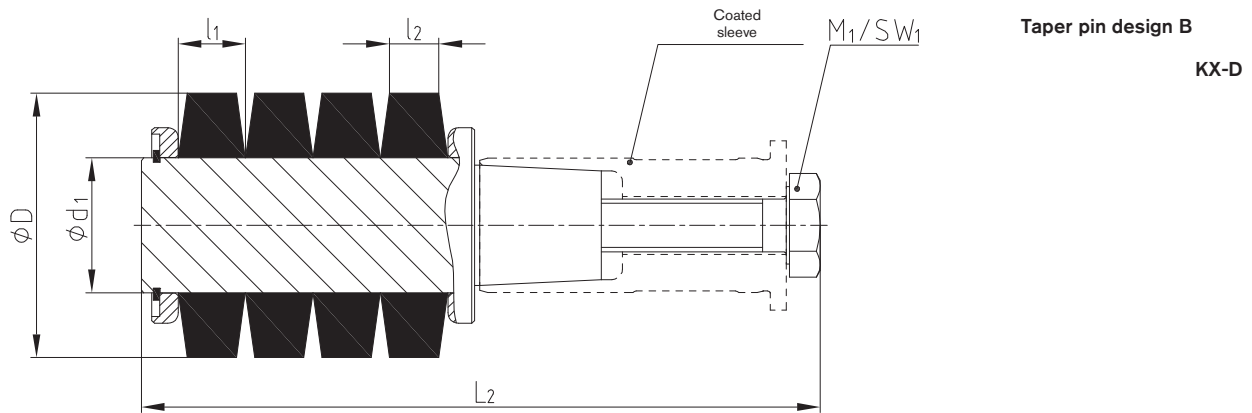


REVOLUX®

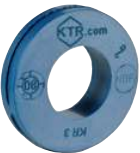
REVOLEX® KX-D

Flexible pin & bush coupling

Technical data of pins



Technical data										
Size	Pin		Elastomer ring NBR 80 Shore A			Pin		Screw DIN EN ISO 4014/4017		Tightening torque T _A [Nm]
	Size	Number								
		KX-D	D	l ₁	l ₂	d ₁	L ₂	M ₁	SW ₁	
75	3	10								
85		12				25.5	129	M10	16	67
95		14								
105		16								
120	4	14								
135		16				30.7	178	M12	18	115
150		18								
170	5	14								
190		16				43.2	220	M16	24	290
215		18								
240		14								
265	6	16								
280		18				58.4	290	M24	36	970
305		20								
330		24								
355	7	16								
370		20				75	393	M30	46	1350
470		22								
520	8	18								
590		20				95	523	M36	55	2250
650		24								

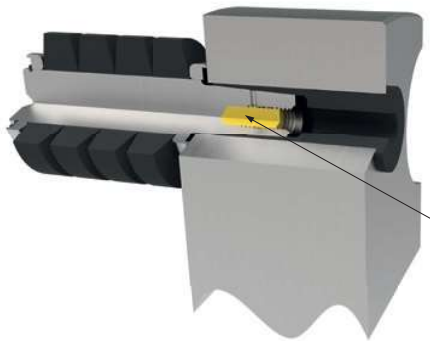
General information on the elastomer rings			
Material	Perbunan [NBR]	Natural rubber [NR]	Perbunan [NBR]
Hardness	80 Shore A	80 Shore A	80 Shore A
Permanent temperature range [°C]	-30 to +80	-50 to +70	-30 to +80
Max. temperature (short time) [°C]	-50 to +120	—	—
Colour	black	black	blue
Operating range	STANDARD	sub-zero temperatures	electrically insulating and backlash-free, e. g. cableway drives
			

Assembly/disassembly



① Unscrew screw

ROTEX®

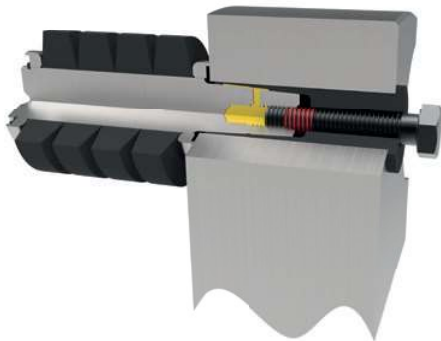


② Seal the screw in this space with sealing tape (e. g. Loctite® 55)

POLY-NORM®

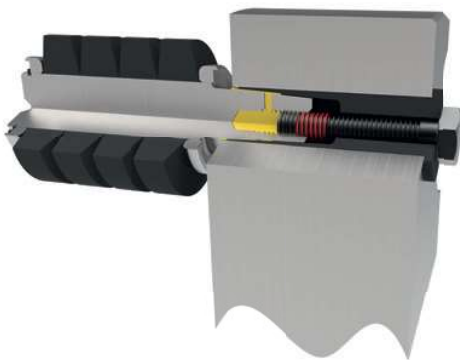


③ Fill the tapped hole with standard grease for 3/4



④ Screw in the screw (no special tools required)

POLY



⑤ The hydraulic pressure is transmitted onto the pin pressing it out of the taper fit.

REVOLUX®